

Xinran Liu

Ph.D. candidate in Computer Science
Vanderbilt University School of Engineering

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Research Interests

Themes Enhancing the efficiency and robustness of Machine Learning methods
Methods Sliced Optimal Transport, Linear Optimal Transport and Partial Optimal Transport
Applications Medical imaging, earth science, point-cloud analysis, shape matching and transfer learning

Education

- 2022 – **Vanderbilt University**, Ph.D. in Computer Science
- o Area: Optimal Transport, Machine Learning
 - o Advisor: Soheil Kolouri
 - o Committee: Soheil Kolouri, Akram Aldroubi, Alan Peters, Catie Chang, Daniel Moyer
- 2019 – 2021 **Vanderbilt University**, M.S. in Mathematics
- 2015 – 2019 **Chongqing University**, B.S. in Mathematics
- o Thesis: Persistent homology and applications in data science

Selected Publications

- 2025 **Liu, X.**, Bai, Y., Lu, Y., Soltoggio, A. and Kolouri, S. Wasserstein task embedding for measuring task similarities. *Neural Networks*, 181, 106796.
- 2024 **Liu, X.**, Bai, Y., Martín, R.D., Shi, K., Shahbazi, A., Landman, B.A., Chang, C. and Kolouri, S. Linear Spherical Sliced Optimal Transport: A Fast Metric for Comparing Spherical Data. In: The Thirteenth International Conference on Learning Representations (ICLR)
- 2024 **Liu, X.**, Martín, R.D., Bai, Y., Shahbazi, A., Thorpe, M., Aldroubi, A. and Kolouri, S. Expected sliced transport plans. In: The Thirteenth International Conference on Learning Representations (ICLR).
- 2024 Lu, Y.*, **Liu, X.***, Soltoggio, A. and Kolouri, S. SLOSH: Set locality sensitive hashing via sliced-wasserstein embeddings. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (pp. 2566-2576).
- 2023 Diaz Martin, R., Medri, I., Bai, Y., **Liu, X.**, Yan, K., Rohde, G.K. and Kolouri, S. LCOT: Linear circular optimal transport. In: The Twelfth International Conference on Learning Representations (ICLR)

Teaching

- 2025 Co-Instructor, **CS 5262: Foundations of Machine Learning**, VU
- 2024 Teaching Assistant, **CS 8395-03: Machine Learning for Dynamical Systems**, VU
- 2023 Teaching Assistant, **CS 8395-02: Statistical Foundations of Deep Learning**, VU
- 2020 – 2021 Teaching Assistant, **MATH 1301: Accelerated Single-Variable Calculus II**, VU

Professional Activities and Service

Reviewer NeurIPS, ICML, ICLR, AAAI

AI scholar Vanderbilt Lab for Immersive AI Translation (VALIANT)

Talks and Tutorials

- ‘Optimal Transport for Practitioners’
 - o AI summer school 2025. VALIANT at Vanderbilt, August 2025
- ‘Optimal Transport in Learning Set-Structured Data’
 - o Women in Data Science Conference. Data Science Institute at Vanderbilt, March 2023

Awards

- 2018 Study Abroad Fellowship (Fully funded award covering tuition and living expenses)
- 2018-2019 Lixin Tang Scholarship (Top 0.2% university-wide)
- 2016 National Scholarship (Top 0.1% nationwide, China)
- 2015-2019 Excellent Academic Performance Scholarship (Every semester, top 3% university-wide)

Professional Training

- July 2025 **Optimal Transport Summer School**, *University of California, Santa Barbara*, US
- June 2022 **Summer School on Optimal Transport**, *Pacific Institute for the Mathematical Sciences*, Seattle, US
- July 2018 **The 1st International Undergraduate Mathematics Summer School**, Peking University, Beijing, China

Professional Memberships

AMS (2019–), IEEE (2022–)

Full List of Publications

Most recent publications via [Google Scholar](#).

Preprints

- 2025 Nath, S., Peridis, C., Benjamin, E., **Liu, X.**, Kolouri, S., Kinnell, P., Li, Z., Liu, C., Dora, S. and Soltoggio, A. Collaborative Learning in Agentic Systems: A Collective AI is Greater Than the Sum of Its Parts. *arXiv preprint arXiv:2506.05577*.
- 2025 NaderiAlizadeh, N., Salehi, D., **Liu, X.** and Kolouri, S. Constrained Sliced Wasserstein Embedding. *arXiv preprint arXiv:2506.02203*.
- 2025 Bai, Y., Tran, H., Du, H., **Liu, X.** and Kolouri, S. Fused Partial Gromov-Wasserstein for Structured Objects. *arXiv preprint arXiv:2502.09934*.

Conference and Journal Publications

- 2025 **Liu, X.**, Bai, Y., Lu, Y., Soltoggio, A., & Kolouri, S. Wasserstein task embedding for measuring task similarities. *Neural Networks*, 181, 106796.
- 2025 Shahbazi, A., Akbari, E., Salehi, D., **Liu, X.**, Naderializadeh, N. and Kolouri, S. ESPFormer: Doubly-Stochastic Attention with Expected Sliced Transport Plans. In: The Forty-Second International Conference on Machine Learning (ICML)
- 2024 Soltoggio, A., Ben-Iwhiwhu, E., Braverman, V., Eaton, E., Epstein, B., Ge, Y., Halperin, L., How, J., Itti, L., Jacobs, M.A. and Kantharaju, P., **Liu, X.** ... A collective AI via lifelong learning and sharing at the edge. *Nature Machine Intelligence*, 6(3), pp.251-264.

- 2024 **Liu, X.**, Bai, Y., Martín, R.D., Shi, K., Shahbazi, A., Landman, B.A., Chang, C. and Kolouri, S. Linear Spherical Sliced Optimal Transport: A Fast Metric for Comparing Spherical Data. In: The Thirteenth International Conference on Learning Representations (ICLR)
- 2024 **Liu, X.**, Martín, R.D., Bai, Y., Shahbazi, A., Thorpe, M., Aldroubi, A. and Kolouri, S. Expected sliced transport plans. In: The Thirteenth International Conference on Learning Representations (ICLR).
- 2024 Bai, Y., Martin, R.D., Kothapalli, A., Du, H., **Liu, X.** and Kolouri, S. Partial Gromov-Wasserstein Metric. In: The Thirteenth International Conference on Learning Representations (ICLR).
- 2024 Tran, H., Bai, Y., Kothapalli, A., Shahbazi, A., **Liu, X.**, Martin, R.P.D. and Kolouri, S. Stereographic Spherical Sliced Wasserstein Distances. In: The Forty-First International Conference on Machine Learning (ICML)
- 2024 **Liu, X.**, Lu, Y., Abbasi, A., Li, M., Mohammadi, J. and Kolouri, S. Teaching networks to solve optimization problems. *IEEE Access*, 12, pp.17102-17113.
- 2024 Lu, Y.*, **Liu, X.***, Soltoggio, A. and Kolouri, S. SlosH: Set locality sensitive hashing via sliced-wasserstein embeddings. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (pp. 2566-2576).
- 2023 Nath, S., Peridis, C., Ben-Iwhiwhu, E., **Liu, X.**, Dora, S., Liu, C., Kolouri, S. and Soltoggio, A. Sharing lifelong reinforcement learning knowledge via modulating masks. In: *Conference on Lifelong Learning Agents* (pp. 936-960). PMLR..
- 2023 Diaz Martin, R., Medri, I., Bai, Y., **Liu, X.**, Yan, K., Rohde, G.K. and Kolouri, S. LCOT: Linear circular optimal transport. In: The Twelfth International Conference on Learning Representations (ICLR)

Refereed Workshop Contributions

- 2024 Kothapalli, A., Shahbazi, A., **Liu, X.**, Sheng, R. and Kolouri, S., Equivariant vs. Invariant Layers: A Comparison of Backbone and Pooling for Point Cloud Classification. In: *ICML 2024 Workshop on Geometry-grounded Representation Learning and Generative Modeling*.
- 2023 **Liu, X.**, Bai, Y., Tran, H., Zhu, Z., Thorpe, M. and Kolouri, S., PTLP: Partial Transport L^p Distances. In: *NeurIPS 2023 Workshop Optimal Transport and Machine Learning*.